

**1510.06699: residuals**

equations measured 2026-09-05T11:24:38Z against the model of 2026-08-21

Flagged, not acted on

| Identifier                                                                                                                                                                             | Page | Conf.              | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1510.06699_<br>E00168 (184c)                                                                                                                                                           | 31   | ■ 0.814<br>● C +25 | $\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b^b_\mu - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, (184b) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger \mu} = 0, (184c) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, (184d) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, (184e) \end{aligned}$                                                             | $\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b^b_\mu - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger \mu} = 0 \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, \end{aligned}$ | $\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b^b_\mu - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, (184b) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger \mu} = 0, (184c) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, (184d) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, (184e) \end{aligned}$ |
| 1510.06699_<br>E00003 (4)                                                                                                                                                              | 5    | ■ 1.000<br>● C +28 | $\begin{aligned} & L_{\mathrm{G}} = \alpha_1 \mathcal{R}^2 + \alpha_2 \mathcal{R}_{ab} \mathcal{R}^{ab} + \alpha_3 \mathcal{R}_{ab} \mathcal{R}^{ba} + \alpha_4 \mathcal{R}_{abcd} \mathcal{R}^{abcd} + \alpha_5 \mathcal{R}_{abcd} \mathcal{R}^{acbd} + \alpha_6 \mathcal{R}_{abcd} \mathcal{R}^{cdab} + \xi \mathcal{H}_{ab} \mathcal{H}^{ab} \equiv L_{\mathcal{R}^2} + L_{\mathcal{H}^2}, \\ & L_{\mathrm{G}} = \alpha_1 \mathcal{R}^2 + \alpha_2 \mathcal{R}_{ab} \mathcal{R}^{ab} + \alpha_3 \mathcal{R}_{ab} \mathcal{R}^{ba} + \alpha_4 \mathcal{R}_{abcd} \mathcal{R}^{abcd} + \alpha_5 \mathcal{R}_{abcd} \mathcal{R}^{acbd} + \alpha_6 \mathcal{R}_{abcd} \mathcal{R}^{cdab} + \xi \mathcal{H}_{ab} \mathcal{H}^{ab} \equiv L_{\mathcal{R}^2} + L_{\mathcal{H}^2}, \end{aligned}$ | $L_{\mathrm{G}} = \alpha_1 \mathcal{R}^2 + \alpha_2 \mathcal{R}_{ab} \mathcal{R}^{ab} + \alpha_3 \mathcal{R}_{ab} \mathcal{R}^{ba} + \alpha_4 \mathcal{R}_{abcd} \mathcal{R}^{abcd} + \alpha_5 \mathcal{R}_{abcd} \mathcal{R}^{acbd} + \alpha_6 \mathcal{R}_{abcd} \mathcal{R}^{cdab} + \xi \mathcal{H}_{ab} \mathcal{H}^{ab} \equiv L_{\mathcal{R}^2} + L_{\mathcal{H}^2},$                                                                                                                                                                                                                                                                                                                        | $L_{\mathrm{G}} = \alpha_1 \mathcal{R}^2 + \alpha_2 \mathcal{R}_{ab} \mathcal{R}^{ab} + \alpha_3 \mathcal{R}_{ab} \mathcal{R}^{ba} + \alpha_4 \mathcal{R}_{abcd} \mathcal{R}^{abcd} + \alpha_5 \mathcal{R}_{abcd} \mathcal{R}^{acbd} + \alpha_6 \mathcal{R}_{abcd} \mathcal{R}^{cdab} + \xi \mathcal{H}_{ab} \mathcal{H}^{ab} \equiv L_{\mathcal{R}^2} + L_{\mathcal{H}^2},$                                                                                                                                                                                                                                                                                                                                                     |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value<br><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured |      |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

55 more flagged rows below the band, stated as a count: 27 component, 28 weak; confidence high 30, mid 17, low 8, none 0

Doubted but correct

| Identifier                                                                                                                                                                             | Page | Conf.             | LaTeX source                                                                     | Rendered                                                                          | Image                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1510.06699_<br>E08241 (A5) [conf<br>0.019]                                                                                                                                             | 47   | ■ 0.019<br>● S -2 | $\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i \gamma^5\right)}{\rho^2+\beta^2}$ | $\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i \gamma^5\right)}{\rho^2+\beta^2}.$ | $\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i \gamma^5\right)}{\rho^2+\beta^2}.$ |
| <b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value<br><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured |      |                   |                                                                                  |                                                                                   |                                                                                   |